| EXP.NO: 7 | Develop an Android Calculator Application using Android Studio |
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## AIM:

To develop an Android application that implements a basic calculator using Android controls such as Button, TextView, and EditText. The calculator will support basic mathematical operations like Addition, Subtraction, Multiplication, and Division.

## ALGORITHM:

1. **Initialize the Android Application:**
   - Set up a new Android project in **Android Studio**.
   - Create a **MainActivity** Java or Kotlin file for the main screen of the application.
2. **Design the User Interface (UI):**
   - Create the following UI components in the **XML layout** file:
     - **EditText** for input fields where the user will enter numbers (two EditTexts for number1 and number2).
     - **Buttons** for each arithmetic operation (Addition, Subtraction, Multiplication, Division).
     - **TextView** to display the result of the operation.
3. **Declare Variables:**
   - Create variables for the **EditText** fields, **Buttons**, and **TextView**.
   - Define variables to hold the input numbers and the result (e.g., num1, num2, result).
4. **Get Input from the User:**
   - Get the values entered by the user in **EditText** fields using getText().toString() and convert them to numerical values (like Integer.parseInt() or Double.parseDouble()).
5. **Set Event Listeners for Buttons:**
   - For each button (Addition, Subtraction, Multiplication, and Division), set an **OnClickListener**.
   - Inside the **onClick()** method for each button, perform the respective arithmetic operation:

- **Addition**: result = num1 + num2
- **Subtraction**: result = num1 - num2
- **Multiplication**: result = num1 * num2
- **Division**: result = num1 / num2 (add validation to prevent division by zero).

6. **Perform Arithmetic Operation:**
   - Check which button was clicked (using view.getId()), and based on the button, perform the respective arithmetic operation.
   - **Handle Division by Zero**: Check if num2 is 0 before performing division and show an appropriate error message if true.
7. **Display the Result:**
   - Use **TextView** to display the result of the operation.
   - Convert the result to a string and set it using setText() method on the **TextView**.
8. **Validate User Inputs:**
   - Before performing any operation, validate the input to ensure the user has entered valid numbers (not empty or non-numeric characters).
9. **Handle Edge Cases:**
   - Ensure the app handles cases where users input invalid values or attempt division by zero. Display appropriate messages for errors.
10. **Finish the Operation:**
    - Once the result is displayed, the user can either perform another calculation or exit the app.

## Code

**Step 1: Create the Project**

- **Template**: Empty Views Activity
- **Project name** – CalculatorApp
- **Package** - com.example.calculatorapp
- **Language**: Java
- **Minimum SDK** – API 24

- **Build system**: Groovy DSL

**Step 2**: Under res → layout → Updated activity_main.xml using **ConstraintLayout**

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <!-- First Number Input -->
    <EditText
        android:id="@+id/editTextNumber1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:inputType="numberDecimal"
        android:hint="Enter first number"
        android:textSize="18sp"
        android:textColorHint="#000000"
        android:typeface="monospace"
        android:textStyle="bold"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_margin="16dp" />

    <!-- Second Number Input -->
    <EditText
        android:id="@+id/editTextNumber2"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:inputType="numberDecimal"
        android:hint="Enter second number"
        android:textSize="18sp"
        android:textColorHint="#000000"
        android:typeface="monospace"
        android:textStyle="bold"
        app:layout_constraintTop_toBottomOf="@id/editTextNumber1"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_margin="16dp" />
```

```
<!-- Row of Operation Buttons -->
<LinearLayout
    android:id="@+id/buttonRow"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:weightSum="4"
    android:layout_margin="16dp"
    app:layout_constraintTop_toBottomOf="@id/editTextNumber2"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent">

    <Button
        android:id="@+id/buttonAdd"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Add" />

    <Button
        android:id="@+id/buttonSubtract"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Subtract" />

    <Button
        android:id="@+id/buttonMultiply"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Multiply" />

    <Button
        android:id="@+id/buttonDivide"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Divide" />
</LinearLayout>

<!-- Clear Button -->
<Button
    android:id="@+id/buttonClear"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Clear"
        app:layout_constraintTop_toBottomOf="@id/buttonRow"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="12dp"/>

    <!-- Result TextView -->
    <TextView
        android:id="@+id/textViewResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result will appear here"
        android:textSize="18sp"
        android:textStyle="bold"
        app:layout_constraintTop_toBottomOf="@id/buttonClear"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="24dp" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

**Step 3** : Update the MainActivity.java file with this content

```
package com.example.calculatorapp;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    EditText editTextNumber1, editTextNumber2;
    Button buttonAdd, buttonSubtract, buttonMultiply, buttonDivide, buttonClear;
    TextView textViewResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

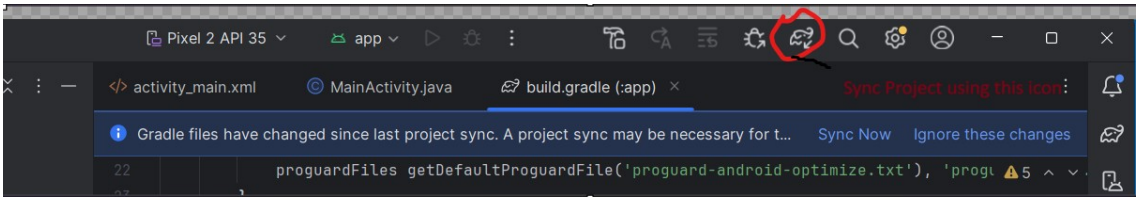
```
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    // Link XML elements with Java code
    editTextNumber1 = findViewById(R.id.editTextNumber1);
    editTextNumber2 = findViewById(R.id.editTextNumber2);
    buttonAdd = findViewById(R.id.buttonAdd);
    buttonSubtract = findViewById(R.id.buttonSubtract);
    buttonMultiply = findViewById(R.id.buttonMultiply);
    buttonDivide = findViewById(R.id.buttonDivide);
    buttonClear = findViewById(R.id.buttonClear);
    textViewResult = findViewById(R.id.textViewResult);

    // Add button logic
    buttonAdd.setOnClickListener(view -> calculate('+'));

    // Subtract button logic
    buttonSubtract.setOnClickListener(view -> calculate('-'));

    // Multiply button logic
    buttonMultiply.setOnClickListener(view -> calculate('*'));

    // Divide button logic
    buttonDivide.setOnClickListener(view -> calculate('/'));

    // Clear button logic
    buttonClear.setOnClickListener(view -> {
        editTextNumber1.setText("");
        editTextNumber2.setText("");
        textViewResult.setText("Result will appear here");
    });
}

private void calculate(char operator) {
    String input1 = editTextNumber1.getText().toString();
    String input2 = editTextNumber2.getText().toString();

    if (input1.isEmpty() || input2.isEmpty()) {
        textViewResult.setText("Please enter both numbers.");
        return;
    }

    try {
        double num1 = Double.parseDouble(input1);
```

```
            double num2 = Double.parseDouble(input2);
            double result = 0;

            switch (operator) {
                case '+':
                    result = num1 + num2;
                    break;
                case '-':
                    result = num1 - num2;
                    break;
                case '*':
                    result = num1 * num2;
                    break;
                case '/':
                    if (num2 == 0) {
                        textViewResult.setText("Cannot divide by zero");
                        return;
                    }
                    result = num1 / num2;
                    break;
            }

            textViewResult.setText("Result: " + result);
        } catch (NumberFormatException e) {
            textViewResult.setText("Invalid input");
        }
    }
}
```

**Step 4** – Under Gradle Scripts section → update the build.gradle(Module:app)

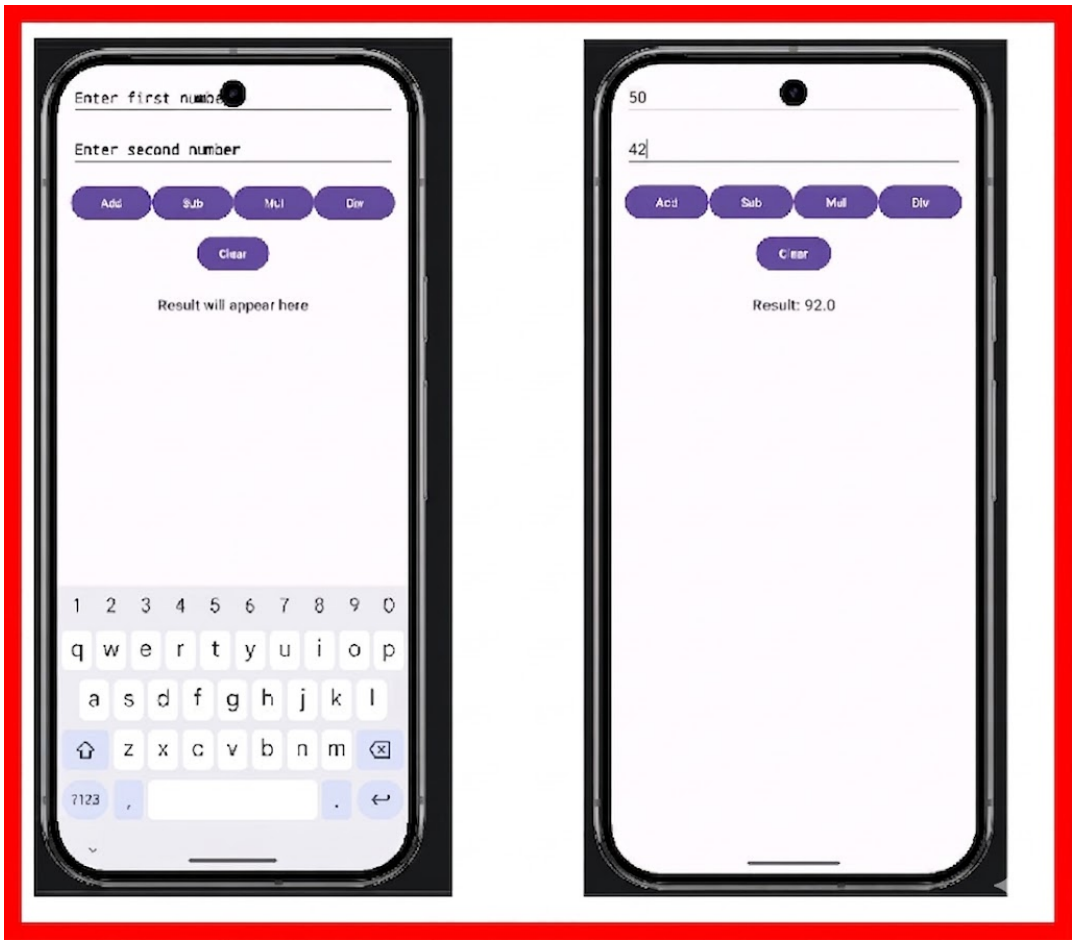
```
plugins {
    alias(libs.plugins.androidApplication)
}

android {
    namespace 'com.example.calculatorapp'
    compileSdk 34

    defaultConfig {
        applicationId "com.example.calculatorapp"
        minSdk 26
        targetSdk 34
        versionCode 1
        versionName "1.0"
```

```
        testInstrumentationRunner "androidx.test.runner.AndroidJUnitRunner"
    }

    buildTypes {
        release {
            minifyEnabled false
            proguardFiles getDefaultProguardFile('proguard-android-optimize.txt'), 'proguard-
rules.pro'
        }
    }
    compileOptions {
        sourceCompatibility JavaVersion.VERSION_1_8
        targetCompatibility JavaVersion.VERSION_1_8
    }
}
dependencies {
    // View/Layout
    implementation 'androidx.constraintlayout:constraintlayout:2.1.4'

    // Core libraries
    implementation 'androidx.core:core:1.12.0'
    implementation libs.appcompat
    implementation libs.material
    implementation libs.activity

    // Testing
    testImplementation libs.junit
    androidTestImplementation libs.ext.junit
    androidTestImplementation libs.espresso.core
}
```




**Step 5** – Click the Play Button to Run the project using the Emulator and verify the output as below

## Output:

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## RESULT:

Thus, An calculator was designed using Android studio which uses controls like Button, TextView, EditText having basic functionality like Addition, Subtraction, multiplication, and Division.